Weekly Progress Report – Week 1

Group: Group G
Project: University Student-Support Case Agent
Course: BSE4104 – Emerging Trends in Software Engineering
Week: Week 1 – Problem Framing and AI-Native Requirements
Week Ending: 7th September 2026

1. Work Completed

During Week 1, the group completed the initial problem framing and requirements activities for the University Student-Support Case Agent. The group confirmed the project problem, target users and primary end-to-end workflow. A Project Charter was developed to define the proposed solution, scope, assumptions, constraints and success criteria. The team also defined the boundaries of the AI system, distinguishing between tasks that can be handled by the AI, deterministic application functions and situations requiring human intervention. Initial user stories and acceptance criteria were prepared, and an initial architecture/context diagram was developed. The project GitHub repository and ClickUp workspace were also established to support collaborative development and progress tracking.

2. Key Engineering Decisions

During Week 1, the group made several decisions to establish clear technical and safety boundaries for the system. The group decided to use publicly available university and course information as the knowledge source for the prototype, while using synthetic student, case and ticket data for functionality that would normally require access to institutional systems. This allows the team to demonstrate realistic workflows without depending on private university systems or exposing real student information.

The group also decided that the agent will be bounded rather than fully autonomous. AI will be used for natural-language understanding, classification, retrieval-based responses and proposing appropriate actions, while deterministic software will handle functions where correctness and authorization must be guaranteed, such as identity validation, case and timetable lookups, access control and ticket routing. Where the AI proposes an action such as creating a support ticket, the application will validate the proposed information before executing it.

The group further established that sensitive or high-impact matters must remain under human authority. Cases involving admissions, grading, disciplinary matters, fee waivers, appeals or changes to official student records will not be autonomously resolved by the agent. The system will either route such cases to human staff or prevent the agent from accessing the relevant functionality entirely. The group also decided that unsupported questions should result in a controlled fallback rather than allowing the model to generate unsupported information, while any stored case memory will be governed by deterministic access, retention and deletion rules.

3. Challenges and Current Response

A key challenge during Week 1 was defining a project that demonstrates meaningful AI and agentic capabilities without depending on access to confidential university systems. The group addressed this by separating public information used for RAG from synthetic data used for case-related tools and demonstrations. Another challenge was keeping the project scope manageable within the eight-week development period. The group responded by defining one primary end-to-end workflow and limiting the agent to approved tools, defined stop conditions and human hand-off situations.

4. Individual Contributions

Member	**Task Owned**	**Evidence**
Ampaire Stacey Nuwagaba – Project & Requirements Lead	Project Charter and primary workflow	Completed Charter attached to ClickUp
Kemirembe Judith – AI Engineering Lead	AI Boundary Matrix	Completed matrix in ClickUp
Kawuma Arnold Fred – Application & Integration Lead	Initial Architecture/Context Diagram	Completed diagram in ClickUp
Aineamatsiko Desmond Cole – Quality & Security Lead	User Stories and Acceptance Criteria	Completed user stories in ClickUp/GitHub
Rachel Mirembe– DevOps & Documentation Lead	GitHub/ClickUp evidence	Repository structure, ClickUp evidence

5. Repository and ClickUp Evidence

GitHub Repository: https://github.com/StaceyAmpaire/Group-G-University-student-support-case-agent
ClickUp Project/List: https://app.clickup.com/1200400000000219/v/o/s/1200400000001102

The GitHub repository contains the project's initial documentation and project structure, while ClickUp contains the Week 1 tasks, assignments, statuses and evidence of individual contributions.

6. Plan for Week 2

In Week 2, the group will integrate and experiment with the selected foundation model and begin prompt engineering. The team will document the model selection, including its capabilities, cost, latency and suitability for the project. A first version of the Prompt Specification will be developed covering the model's role, task, context, constraints, output format and failure behaviour. The team will then establish an initial system prompt and evaluate it using at least ten test cases by comparing expected and actual outputs. At least two meaningful prompt iterations will be documented based on the evaluation results before the group progresses to the RAG implementation in Week 3.